

Snehal Nakhawa

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SUMMARY

Senior Software Engineer with 10 years of experience with proven expertise in building secure, scalable, and cloud-native systems, specializing in application security and open-source dependency analysis. Strong background in distributed systems, multithreaded workflows, and high-performance artifact processing using Java. Recently integrated LLM-based tools into workflows for Agentic AI using classifier. Seeking advance in backend and platform engineering roles focused on secure system design, developer tooling, and large-scale distributed architectures.

PROFESSIONAL EXPERIENCE

Ness Digital Engineering | Engineering Lead | Pearson VUE | Mumbai, India

Oct 2022 - Sept 2025

- Drove development and optimization of RESTful API using Spring boot, improving response latency by 30% throughout efficient **data structures, caching strategies** and streamlined request handling.
- Design and implemented scalable request batching and processing mechanisms, enabling the system to handle high-throughput traffic with minimal latency and improved stability under load.
- Developed **distributed integrations** with external platforms using web client (**Salesforce, Kibo**), **ensuring 99.9% data consistency and fault tolerance across services**.
- Developed a scalable Angular application for efficient API consumption and low-latency performance. Achieved **98% success rate in building and testing performance** of the application built using angular 17 for Mind hub services.
- Deployed services using Containerized with Docker and **deployed on Kubernetes** to support high-throughput and concurrent users.
- Optimized database performance through scripts, stored procedures, and query tuning, enabling efficient large-scale data processing. Improving the application performance by few seconds. **Reducing query latency by 50%**, improving efficiency and scalability of the database.

LTIMindtree | Senior Software Engineer | NEXI Group | Mumbai, India

June 2019 – Sept 2022

- Developed and scaled an API Market place platform for BSF services, enabling customers to securely discover and consume APIs published via WSO2 API Gateway.
- Automated customer onboarding workflows by developing RESTful services for **API key/secret generation and token-based access control**, reducing manual provisioning effort and improving time-to-access.
- Architected event-driven integrations using **RabbitMQ**, enhancing system reliability and enabling scalable asynchronous communication between distributed services.
- Migrated event-driven messaging infrastructure from **RabbitMQ to Apache Kafka**, re-architect topic design, consumer groups, and offset management to support interservice communications and logs streams.
- Built an Angular web app supporting **1K+ users and improving onboarding experience for new users**, with **90%+ test coverage**, and deployed via Docker containers, reducing deployment inconsistencies and improving release efficiency.
- Led onboarding of **25 Customer**, aligned **cross-functional teams** across **engineering, compliance**, and deployment teams, tracked dependencies, and ensured delivery within timelines.

Mastek Limited | Software Engineer | Morrisons | Hitachi Capital (UK) | Mumbai, India

April 2016– May 2019

- Architected and delivered a **real-time digital pricing platform** for retail stores, enabling automated updates of electronic shelf labels and **supporting 1K+ price updates per day** with high accuracy and reliability.
- Engineered high-performance microservices using **Spring Boot**, reducing API response latency by **25–30%** and improving system responsiveness for real-time pricing workflows.
- Developed scalable RESTful APIs integrated with Angular frontend, supporting **5K+ concurrent users** with low-latency interactions and seamless user experience.
- Designed and implemented a **cloud-native, event-driven architecture** using AWS (S3, Lambda, API Gateway, SNS), improving system scalability, fault tolerance, and asynchronous processing efficiency.
- Containerized services using Docker and deployed on **AWS ECS**, enabling automated deployments and improving release efficiency by **40%** while ensuring high availability.
- Containerized microservices using **Docker** and orchestrated deployments with **Kubernetes**, while implementing end-to-end **CI/CD pipelines (Jenkins/GitHub Actions)** to automate build, test, and deployment workflows; enabled **auto-scaling, rolling deployments, and zero-downtime releases**, reducing release time by **40%**.
- Integrated **95%+ test coverage** (unit + integration) into CI/CD pipelines, enabling automated validation, faster releases, and improved deployment confidence.

Trimax private Limited | Java Engineer | Siddhivinayak| MSRTC |Mumbai, India

January 2015– April 2016

- Architected and delivered a full-stack temple management web application enabling devotees to browse, book, and pay for puja time slots online — eliminating manual booking queues and improving user accessibility.
- Designed and implemented a **slot booking and scheduling engine** allowing users to select preferred date, time, and puja type with real-time availability checks.
- Integrated a third-party **Payment Gateway** into the temple booking platform, enabling secure online transactions with end-to-end payment flow handling — achieving a **99.9% successful transaction rate** and ensuring a seamless, trustworthy checkout experience for devotees.
- Deployed the application on **Apache Tomcat** web server, ensuring stable production environment and reliable uptime for high-traffic religious event periods.

EDUCATION

Master of Science, Computer Science

Expected May 2027

Stevens Institute of Technology, Hoboken, NJ

SKILLS

Java, Python, JavaScript, TypeScript, Spring Boot, Spring Security, Hibernate, Angular, Web-client, REST-API, Micro-Services, GraphQL, RMQ, Kafka, MVC, AWS (EC2, Lambda, API Gateway, ECS), Redis, Load Balancing, Docker, Terraform, Kubernetes, CI/CD, GitHub, MySQL, DynamoDB, SQL, NoSQL, Splunk, Maven, Gradle, Junit, Mockito, Karma. Data Structures & Algorithms, OOP, Microservices, System Design, Machine Learning Models, Reinforcement Learning, DQN, TensorFlow, Agentic RAG, LLM, Cursor, Copilot, Critical Thinking, Written and Verbal Communication, Ownership, Articulate, Innovative, Growth Mindset, Influence, Business Analysis, Best Practices, Decision Making, Problem Solving, Opportunities, Leadership Skills, Resolving Issues, Customer Obsessed, Stakeholder Engagement, Software Engineering, Prioritization, Process Development, Quality, Product Development.

ACADEMIC PROJECTS — STEVENS INSTITUTE OF TECHNOLOGY

Enterprise Software Architecture and Design

Deployed cloud-native application using **Quarkus** and **Kubernetes**, leveraging Quarkus fast startup and low memory footprint for containerized microservices — integrating **MCP (Model Context Protocol) server** for AI-powered tooling and context management enabling scalable, low-latency deployments.

Industrial IoT Stimulation

Built a **predictive maintenance platform** using Random Forest and Isolation Forest on sensor streams, enabling early fault detection **and reducing potential downtime by 80%**.

Developed a **real-time observability dashboard** (Streamlit) to monitor machine health and model outputs at scale.

Integrated an **LLM-powered agent (Ollama + LangChain)** to deliver conversational diagnostics and automated insights, improving accessibility of ML predictions across teams.

Reinforcement Learning Lunar Lander (DQN)

Implemented Deep Q-network DQN agent using TensorFlow and PyGame. Using reward strategy. Finetuning the model to get the max reward.

Trained agent using Frame-by-Frame learning and reward shaping.

Evaluated performance using episodic reward and convergence metrics.

AI for Software Testing

Analyzed testing behavior for AI coding agents using AIDev datasets. • Used tools like PMD, EMAssist, Understand for coding quality and refactoring. • Studied difference between AI generated and Human generated test cases.

CERTIFICATE

AWS – Developer Associate | Amazon Web Services